Chat Export for sanidhya.rana_contractor@celanese.com

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: fuse switch for electrical device

Assistant: To help identify the best material for your fuse switch application, could you please clarify:

1. What is the exact use of the polymer in your fuse switch? For example, is it for the housing, lever, cover, retainer, or another component?
2. Where will this fuse switch be used? Is it for consumer electronics, industrial control panels, automotive systems, or another end-use scenario?

Your answers will help me understand the requirements more precisely and suggest suitable material options.

User: a back cover and an internal part for industrial control panel

Assistant: Thank you for clarifying the application. To proceed, could you please provide:

1. What are the typical operating conditions for your industrial control panel? For example, is it exposed to high temperatures, vibration, dust, or humidity? (Based on your application, these panels are usually indoors and protected, but may face moderate heat and vibration.)
2. What is the minimum wall thickness required for the back cover or internal part?
3. Do you have any preferred base materials for these components (such as specific

polymers you've used before)?

4. Are you open to alternative material suggestions if they meet your requirements?

Your answers will help ensure the material suggestions are tailored to your needs.

User: dont have preference and I am open to material suggestions.

Assistant: Here are the key material requirements (CTQs) assumed for your application:

1. Minimum wall thickness: 1.5 mm (typical for rigid covers and internal frames)
2. Flammability classification: UL94 V-0 at 1.5 mm
3. Flame retardant (halogen-free preferred)
4. Injection molding required
5. Dimensional stability: Low dimensional change (target within 0–0.594%)
6. Dielectric strength: Medium band (23–39 kV/mm)
7. Volume resistivity: High insulation (8.25e15–1.65e16 Ohm·m)

Are all of these requirements correct for your application? Please confirm or let me know if any need adjustment.

User: looks good recommend the grades

Assistant: Here are potential material options for your back cover and internal part for a fuse switch in an industrial control panel:

1. FRIANYL® A3 RV0 NC 1102 (natural color, PA66, halogen-free, UL94 V-0 at 1.5 mm, high insulation, low dimensional change)
2. FRIANYL® A3 RV0 (natural color, PA66, same performance as above)
3. FRIANYL® A3 RV0 GY 7035/P (grey color, PA66, same performance as above)
4. Zytel® FR7025V0F NC010 (natural color, PA66, halogen-free, UL94 V-0 at 1.5 mm,

widely used in E&E)

5. CELANEX® XFR 6852 GF30 (halogen-free FR PBT, high insulation, high rigidity, UL94 V-0 at 1.5 mm)

These grades may be suitable for your application based on technical requirements. Please review the options and consider your processing and color preferences. Final material selection and product design decisions should be validated for your specific use case. If your application involves critical safety or regulatory compliance, ensure compliance with all applicable standards and regulations. Celanese does not manufacture or certify products for these applications.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.